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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Euthanasia Agents: 69 Mice were sacrificed by cervical dislocation, and inguinal (subcutaneous) fat pads were aseptically removed and collected into a 50-mL Falcon tube containing sterile ice-cold Hanks balanced salt solution with Ca2+ and Mg2+ (HBSS) (Gibco, Waltham, MA, USA) and 50 UmL penicillin-streptomycin (PenStrep) (Lonza, Basel, Switzerland).
IACUC: Animal experiments All experiments involving mice were conducted in accordance with the Animals (Scientific Procedures) Act 1986 under a UK Home Office project license approved by the Sheffield University Ethical Review Committee Project Applications and Amendments Sub-Committee and by the UK Animal Procedures Committee (London, UK).
IRB: Reprogramming of human fibroblasts into neural progenitor cells (iNPCs) Skin biopsy collection from ALS patients and healthy donors (Table S1) was approved by the Research Ethics Committee of Sheffield (study number STH16573, reference 12YH0330) or by the Institutional Review Board of the Ohio State University (ethics number IRB08-00402) after informed consent was obtained.
Consent: Reprogramming of human fibroblasts into neural progenitor cells (iNPCs) Skin biopsy collection from ALS patients and healthy donors (Table S1) was approved by the Research Ethics Committee of Sheffield (study number STH16573, reference 12YH0330) or by the Institutional Review Board of the Ohio State University (ethics number IRB08-00402) after informed consent was obtained.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
Materials and methods Isolation and expansion of mouse adipose-derived stem cells mADSCs were isolated from 8- to 10-week-old female C57BL6J mice as previously described with minor modifications.
Subject Demographics
Age: Materials and methods Isolation and expansion of mouse adipose-derived stem cells mADSCs were isolated from 8- to 10-week-old female C57BL6J mice as previously described with minor modifications.
Randomization

not detected.

Blinding

All behavioral tests were performed by the same investigator, who was blinded to the treatment group.

Power Analysis

For the in vivo therapeutic study, we followed the recommendations for the conduct of pre-clinical studies in the SOD1G93A mouse model released by the ALS Therapy Development Institute (ALS TDI), which were successively updated in 2009 following an international conference held in Germany, 11-13 June 2009. 74 To determine group size, a statistical power analysis using G*Power 3.1.

Replication

Passage 4 mAstrocytes from symptomatic (90 day old) SOD1G93A or NTg age-matched control mice were seeded in triplicate (technical replicate) onto 1 well of a 24-well plate previously coated with 1:1,000 poly-l-ornithine (Sigma, St. Louis, MO, USA) and laminin 1:200 (Sigma, St. Louis, MO, USA), at a concentration of 50,000 cellswell. mAstrocytes were maintained for 24 h in astrocyte basal medium (DMEM, 10 FBS, 1 PenStrep).

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Cell Line Authentication

Authentication: Nissl-positive a-MNs were identified and counted depending on their topography, morphology, and size.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
The following antibodies were used: mouse anti-CD90-phycoerythrin (PE) (0.15 mg5 x 105 cells), mouse anti-CD44-allophycocyanin (APC) (0.13 mg5 x 105 cells), mouse anti-CD106-phycoerythrin-cyanine7 (PE-Cy7) (0.20 mg5 x 105 cells), mouse anti-CD45-fluorescein isothiocyanate (FITC) (0.15 mg5 x 105 cells), mouse anti-CD34-PE (0.13 mg5 x 105 cells), mouse anti-CD31-APC (0.15 mg5 x 105 cells), mouse anti-CD29-PE-Cy7 (0.50 mg5 x 105 cells), mouse anti-CD105-PE-Cy7 (0.50 mg5 x 105 cells), and mouse anti-CD11b-FITC (0.13 mg5 x 105 cells).	anti-CD90-phycoerythrin		
	anti-CD44-allophycocyanin		
	anti-CD106-phycoerythrin-cyanine7		
	PE-Cy7		
	anti-CD45-fluorescein		
	anti-CD34-PE		
	anti-CD31-APC		
	anti-CD29-PE-Cy7		
	anti-CD105-PE-Cy7		
	anti-CD11b-FITC		
Experimental Models: Cell Lines			
One day before transfection, HEK293T cells were plated onto 20 Petri dishes at a seeding density of 3 x 106 cellsplate and cultured in DMEM, 10 FBS, and 1 PenStrep.	HEK293T		Suggestion: RRID:CVCL_0063(link)
TU was calculated by quantifying the number of EGFP-positive HeLa cells after transduction with serial dilutions (10-2, 10-3, and 10-4 mL of vector) of the concentrated lentivirus preparation.	HeLa		Suggestion: RRID:CVCL_0030(link)
Experimental Models: Organisms/Strains			
Materials and methods Isolation and expansion of mouse adipose-derived stem cells mADSCs were isolated from 8- to 10-week-old female C57BL6J mice as previously described with minor modifications.	C57BL6J		Suggestion: RRID:IMSR_JAX:000664(link)
Transgenic C57BL6J-Tg (SOD1G93A)1GurJ mice of the defined C57BL6J-OlaHsd inbred genetic background were used in this study.	C57BL6J-Tg (SOD1G93A)1GurJ		
	C57BL6J-OlaHsd		
Mouse astrocyte and ADSC separated co-cultures mAstrocytes derived from	NTg		

SOD1G93A and age-matched NTg mice obtained from three separate preparations were plated onto 24-well plates coated with 1:1,000 poly-l-ornithine (A-004-C, Sigma-Aldrich) at a concentration of 50,000 cells per well.			
Recombinant DNA			
The pLV-EGFP plasmid is a HIV-1-based expression vector in which the transcription of EGFP is driven by the mouse phosphoglycerate kinase (mPGK) promoter.	pLV-EGFP		
Lentivirus was produced from co-transfection of HEK293T cells with the transfer plasmid pLV-SIN-EGFP and the three packaging plasmids pCMVD8.92, pRSV-Rev, and pMDG2.	pCMVD8.92		
	pRSV-Rev		
	pMDG2		
Each plate was transfected with 13 mg of pLV-SIN-EGFP, 13 mg of pCMVD8.92, 3 mg of pRSV-Rev, and 3.75 mg of pMDG.	pLV-SIN-EGFP		
	pMDG		
Software and Algorithms			
The tissue was digested with 0.075 collagenase I (Sigma, St. Louis, MO, USA) and 2 bovine serum albumin (BSA) (Sigma, St. Louis, MO, USA) at 37degC in a water bath for 60-80 min with gentle shaking every 10 min.	Sigma		
Cells were then filtered, washed in HBSS, and cultured in Dulbeccos modified Eagles medium (DMEM) (Lonza, Basel, Switzerland) containing 4,500 mgL glucose, l-glutamine, 10 fetal bovine serum (FBS) (MSC-Qualified, 13662-011, Gibco, Waltham, MA, USA), and 1 PenStrep.	Gibco		
Stained cells were acquired on a BD LSRII FACS instrument (BD Biosciences, Franklin Lakes, NJ, USA) and analyzed with FlowJo 7.6.1 (FlowJo LLC, Ashland, OR, USA).	FlowJo		Suggestion: FlowJo, RRID:SCR_008520(link)
For the in vivo therapeutic study, we followed the recommendations for the conduct of pre-clinical studies in the SOD1G93A mouse model released by the ALS Therapy Development Institute (ALS TDI), which were successively updated in 2009 following an international conference held in Germany, 11-13	G*Power		Suggestion: G*Power, RRID:SCR_013726(link)

June 2009. 74 To determine group size, a statistical power analysis using G*Power 3.1.			
The number of viable MNs was counted with the Columbus Data Storage and Analysis System (RRID:SCR_007149 Perkin Elmer, Waltham, MA, USA).	Analysis		Cellular Imaging and Analysis (RRID:SCR_007149)(link)
Multiplex bead-based immunoassay Soluble cytokines in supernatants were measured with the use of the BD Cytometric Bead Array (CBA) Flex Set multiplexed bead-based immunoassay (BD Biosciences), which allows up to 30 analytes to be measured simultaneously.	BD Biosciences		
The data files were collected as Excel sheets and analyzed with GraphPad 7.	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)
Statistical analysis All statistical analyses were performed with GraphPad Prism 7 (GraphPad, San Diego, CA, USA).	GraphPad Prism		Suggestion: GraphPad Prism (RRID:SCR_002798)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
A power of 80 was used with a two-tailed Students t test (a = 0.05, b = 0.8).	t test
The 1-way ANOVA with Tukeys multiple comparison test, 2-way ANOVA with Sidaks multiple comparison test, 2-way ANOVA with Bonferroni post hoc test, Students t test, or Mann-Whitney U test was used depending on the number of variables in each experiment.	ANOVA
	Tukeys multiple comparison test
	Sidaks multiple comparison test
	Bonferroni post hoc test
	Students t test
	Mann-Whitney U test

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